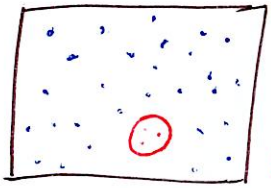
"Derivation" of the Boltzmann Distribution



Universe with Energy E_0 such that
Heat Bath with Energy E_B $E_0 = E_B + \varepsilon$
Microstate with Energy ε

Assume: microstate and bath have reached thermal eq. and the bath is much larger than the ~~micro~~ microstate

$$E_B \gg \varepsilon$$

The probability of observing microstate i is related to the number of ways we can arrange the bath around the microstate. That is,

$$P(\varepsilon_i) \propto \Omega_B \equiv \Omega(E_0 - \varepsilon_i)$$

Entropy is defined as: $S = k_B \ln(\Omega)$. This is mathematically advantageous, as we like to <u>multiply probabilities</u>* but <u>add energies</u>*.

extra worksheet/appendix

* Can you think of examples of why this is?

Now we can substitute our terms:

$$S = k_B \ln\left[\Omega(E_0 - \varepsilon_i)\right]$$

and we assumed that $E_0 \gg \varepsilon_i$, which allows us to Taylor Expand our expression

$$S(E_0 - \varepsilon_i) \approx S(E_0) + \left.\frac{\partial S}{\partial E}\right|_{E=E_0}(-\varepsilon_i) + \frac{1}{2}\left.\frac{\partial^2 S}{\partial E^2}\right|_{E=E_0}(-\varepsilon_i^2) + \dots$$

<u>or</u>

$$S(E_0) - \varepsilon_i \left.\frac{\partial S}{\partial E}\right|_{E_0} - \frac{\varepsilon_i^2}{2}\left.\frac{\partial^2 S}{\partial E^2}\right|_{E_0} + \dots$$

but $\partial^2 S$ scales as the size of the universe and ∂E^2 scales as size of universe squared. Thus, for sufficiently large universe, $\frac{\partial^n S}{\partial E^n} = 0$ for all $n > 1$

This means we can drop out all Taylor Expansion terms after the first derivative and we get

$$S = S(E_0) - \varepsilon_i \left.\frac{\partial S}{\partial E}\right|_{E_0}$$

Now we return to our definition of entropy: $S = k_B \ln(\Omega)$ and solve for Ω:

$\Omega(E_0 - \varepsilon_i) = e^{S/k_B}$. Plug our Taylor expansion in for S

$$\Omega(E_0 - \varepsilon_i) = e^{\left(\frac{S(E_0)}{k_B} - \frac{\varepsilon_i \left.\frac{\partial S}{\partial E}\right|_{E_0}}{k_B}\right)}$$

From thermodynamics: $\left.\frac{\partial S}{\partial E}\right|_{E_0} = \frac{1}{T_0}$ (temp of universe)

Thus:

$$\Omega(E_0 - \varepsilon_i) = e^{\left(\frac{S(E_0)}{k_B} - \frac{\varepsilon_i}{k_B T}\right)} = \underbrace{e^{\left(\frac{S(E_0)}{k_B}\right)}}_{\text{This term is a constant "C"}} \cdot e^{\left(-\frac{\varepsilon_i}{k_B T}\right)}$$

$$\Rightarrow \boxed{P_i = C e^{\left(\frac{-\varepsilon_i}{k_B T}\right)}} \; *$$

* What happens when we ask questions about relative probabilities of two states?

Probability is normalized such that $\sum_i P_i = 1$

$$\therefore \sum C e^{(-\varepsilon_i / k_B T)} = 1 \rightarrow C = \frac{1}{\sum_i e^{(-\varepsilon_i / k_B T)}} \text{ and}$$

$$P_i = \frac{e^{(-\varepsilon_i / k_B T)}}{\sum_i e^{(-\varepsilon_i / k_B T)}} = \frac{e^{(-\varepsilon_i / k_B T)}}{Q(N,V,T)}$$

where $Q(N,V,T)$ is the <u>partition function</u>